

05 — Features / Use Cases (numbered)

Every feature has a stable number **Fxx**. The same numbers appear **in the source code** as comments (each Lambda has a **Feature: Fxx** docstring; each frontend page has a **<!-- ... Features: Fxx -->** comment; **api.js** lists the features it covers). This satisfies the requirement to number each feature and annotate, in the code, which feature(s) each file implements.

Actors / permission groups

- **Guest** — unauthenticated invitee (uses a personal RSVP link only).
- **Host** (Cognito group **host**) — authenticated event organizer; manages **their own** event. New sign-ups are auto-added to this group by a Post-Confirmation trigger.
- **Admin** (Cognito group **admin**) — platform administrator; manages **all** events.

Feature catalog

#	Feature	Actor	Frontend	Backend (route → Lambda)
F01	Sign up (email + password)	Host	signup.html	Cognito SignUp
F02	Confirm email & resend code	Host	signup.html	Cognito ConfirmSignUp / ResendConfirmationCode
F03	Sign in	Host/Admin	login.html	Cognito InitiateAuth
F04	Reset forgotten password	Host	forgot.html	Cognito ForgotPassword / ConfirmForgotPassword
F05	Sign out / session management	Host/Admin	all authed pages, api.js	client-side token clear
F06	Create event	Host	create-host.html	POST /hosts → add_host
F07	View event dashboard	Host	host.html	GET /hosts → get_host
F08	Edit event details & categories	Host	host.html	PUT /hosts → update_host
F09	Delete event	Host	host.html	DELETE /hosts → delete_host
F10	Define tables & capacities	Host	host.html	POST /tables → set_tables
F11	Add guest	Host	host.html	POST /guests → add_guest
F12	List guests	Host	host.html	GET /guests → get_guests

#	Feature	Actor	Frontend	Backend (route → Lambda)
F13	Update guest / assign seat	Host	host.html, guest.html	PUT /guests → update_guest
F14	Delete guest	Host	host.html, guest.html	DELETE /guests → delete_guest
F15	Guest RSVP via public link	Guest	rsvp.html	POST /guests/rsvp → rsvp_guest
F16	Automatic seating generation	Host	host.html	POST /seating → generate_seating
F17	Send invitations (one / all)	Host	host.html	POST /invitations/send → send_invitation
F18	Admin: list & manage all events	Admin	admin.html	GET /admin/hosts → list_hosts
F19	Shareable no-login preview / demo link	Host (shares)	host.html?host=, guest.html?host=	reuses get_host / get_guests
F20	Auto-assign new sign-ups to the host group	System (Cognito trigger)	—	auth_post_confirmation (Post-Confirmation)

Use cases (detail)

UC-F01/F02 — Register a host account

1. Visitor opens **signup.html**, enters name, email, password.
2. Cognito creates the user and emails a 6-digit code.
3. Visitor enters the code (or requests a resend) to confirm the account.

UC-F03/F05 — Sign in and out

1. Host enters email + password on **login.html**.
2. On success, tokens are stored in **localStorage** (**seatme_auth**).
3. **Admins** are routed to **admin.html**; hosts to **host.html** (or **create-host.html** if they have no event yet).
4. *Log out* clears the stored tokens.

UC-F04 — Reset a forgotten password

1. Host requests a code on **forgot.html**.
2. Cognito emails a code; host enters it with a new password.

UC-F06 — Create an event

1. After first sign-in the host fills in event name, date, location.
2. `add_host` creates the item with default categories and empty guests/tables.

UC-F07/F08/F09 — Manage the event

- View the dashboard (`get_host`), edit details or categories (`update_host`), or delete the event and all its guests (`delete_host`).

UC-F10/F16 — Tables and seating

1. Host defines tables and capacities (`set_tables`).
2. Host clicks **Generate seating**; `generate_seating` seats every `yes` guest, keeping categories together, and reports who couldn't be seated.

UC-F11/F12/F13/F14 — Guest management

- Add (`add_guest`), list (`get_guests`), update or manually seat (`update_guest`), delete (`delete_guest`).

UC-F15 — Guest RSVP (no login)

1. Guest opens their personal link (`rsvp.html?...`).
2. Chooses *yes/no*, sets party size, optional song; `rsvp_guest` saves it.
3. If seated, the page shows their table number.

UC-F17 — Invitations

- Host emails one guest or the whole list (`send_invitation` via SNS). First contact is a one-time *Confirm subscription* email; afterwards links are delivered directly.

UC-F18 — Admin oversight (second permission group)

1. Admin signs in and lands on `admin.html`.
2. `list_hosts` validates the admin's Cognito **access token** and `admin` group membership, then returns **all** events.
3. Admin can **Open** any event (reusing the host dashboard) or **Delete** it.

UC-F19 — No-login preview / demo link

- A host (or grader) can open `host.html?host=<email>` to view an event **read-only** without signing in — the page shows a read-only banner and disables editing, and any write is rejected server-side. Signing in as the host or an admin unlocks editing. This powers the shareable demo link printed by `redeploy_all.py`.